Hitters Notebook

Code

The hitters dataset consists of 322 observations of 21 variables with the following information - X, AtBat, Hits, HmRun (home runs), Runs, RBI, Walks, Years, CAtBat, CHits, CHmRun, CRuns, CRBI, CWalks, League, Division, PutOuts, Assists, Errors, Salary, New League. The variable definitions are:

X: Name

AtBat: Number of times at bat in 1986

Hits: Number of hits in 1986

HmRun: Number of home runs in 1986

Runs: Number of runs in 1986

RBI: Number of runs batted in in 1986

Walks: Number of walks in 1986

Years: Number of years in the major leagues

CAtBat: Number of times at bat during his career

CHits: Number of hits during his career

CHmRun: Number of home runs during his career

CRuns: Number of runs during his career

CRBI: Number of runs batted in during his career

CWalks: Number of walks during his career

League: A and N indicating player’s league at the end of 1986

Division: E and W indicating player’s division at the end of 1986

PutOuts: Number of put outs in 1986

Assists: Number of assists in 1986

Errors: Number of errors in 1986

Salary: 1987 annual salary on opening day in thousands of dollars

NewLeague: A and N indicating player’s league at the beginning of 1987.

Here League, Division and NewLeagues are variables with 2 categories. We drop rows with missing entries and are left with 263 observations.

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```
rm(list=ls())
hitters <- read.csv("hitters.csv")
str(hitters)
```

```
'data.frame':   322 obs. of  21 variables:
 $ X         : chr  "-Andy Allanson" "-Alan Ashby" "-Alvin Davis" "-Andre Dawson" ...
 $ AtBat     : int  293 315 479 496 321 594 185 298 323 401 ...
 $ Hits      : int  66 81 130 141 87 169 37 73 81 92 ...
 $ HmRun     : int  1 7 18 20 10 4 1 0 6 17 ...
 $ Runs      : int  30 24 66 65 39 74 23 24 26 49 ...
 $ RBI       : int  29 38 72 78 42 51 8 24 32 66 ...
 $ Walks     : int  14 39 76 37 30 35 21 7 8 65 ...
 $ Years     : int  1 14 3 11 2 11 2 3 2 13 ...
 $ CAtBat    : int  293 3449 1624 5628 396 4408 214 509 341 5206 ...
 $ CHits     : int  66 835 457 1575 101 1133 42 108 86 1332 ...
 $ CHmRun    : int  1 69 63 225 12 19 1 0 6 253 ...
 $ CRuns     : int  30 321 224 828 48 501 30 41 32 784 ...
 $ CRBI      : int  29 414 266 838 46 336 9 37 34 890 ...
 $ CWalks    : int  14 375 263 354 33 194 24 12 8 866 ...
 $ League    : chr  "A" "N" "A" "N" ...
 $ Division  : chr  "E" "W" "W" "E" ...
 $ PutOuts   : int  446 632 880 200 805 282 76 121 143 0 ...
 $ Assists   : int  33 43 82 11 40 421 127 283 290 0 ...
 $ Errors    : int  20 10 14 3 4 25 7 9 19 0 ...
 $ Salary    : num  NA 475 480 500 91.5 750 70 100 75 1100 ...
 $ NewLeague: chr  "A" "N" "A" "N" ...
```

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```
hitters<-na.omit(hitters)
str(hitters)
```

```
'data.frame':   263 obs. of  21 variables:
 $ X         : chr  "-Alan Ashby" "-Alvin Davis" "-Andre Dawson" "-Andres Galarraga" ...
 $ AtBat     : int  315 479 496 321 594 185 298 323 401 574 ...
 $ Hits      : int  81 130 141 87 169 37 73 81 92 159 ...
 $ HmRun     : int  7 18 20 10 4 1 0 6 17 21 ...
 $ Runs      : int  24 66 65 39 74 23 24 26 49 107 ...
 $ RBI       : int  38 72 78 42 51 8 24 32 66 75 ...
 $ Walks     : int  39 76 37 30 35 21 7 8 65 59 ...
 $ Years     : int  14 3 11 2 11 2 3 2 13 10 ...
 $ CAtBat    : int  3449 1624 5628 396 4408 214 509 341 5206 4631 ...
 $ CHits     : int  835 457 1575 101 1133 42 108 86 1332 1300 ...
 $ CHmRun    : int  69 63 225 12 19 1 0 6 253 90 ...
 $ CRuns     : int  321 224 828 48 501 30 41 32 784 702 ...
 $ CRBI      : int  414 266 838 46 336 9 37 34 890 504 ...
 $ CWalks    : int  375 263 354 33 194 24 12 8 866 488 ...
 $ League    : chr  "N" "A" "N" "N" ...
 $ Division  : chr  "W" "W" "E" "E" ...
 $ PutOuts   : int  632 880 200 805 282 76 121 143 0 238 ...
 $ Assists   : int  43 82 11 40 421 127 283 290 0 445 ...
 $ Errors    : int  10 14 3 4 25 7 9 19 0 22 ...
 $ Salary    : num  475 480 500 91.5 750 ...
 $ NewLeague: chr  "N" "A" "N" "N" ...
 - attr(*, "na.action")= 'omit' Named int [1:59] 1 16 19 23 31 33 37 39 40 42 ...
  ..- attr(*, "names")= chr [1:59] "1" "16" "19" "23" ...
```

Subset selection

The leaps package in R does subset selection with the regsubsets function. By default, the maximum number of subsets, this function uses is 8. We extend this to do a complete subset selection by changing the default value of nvmax argument in this function. Note that CRBI is in the model with 1 to 6 variables but not in the model with 7 and 8 variables. In the display “*” indicates variable is included and ” ” indicates variable is not included.

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```
#install.packages("leaps")
library(leaps)
?regsubsets
hitters <- hitters[,2:21]
model1 <- regsubsets(Salary~.,data=hitters)
summary(model1)
```

```
Subset selection object
Call: regsubsets.formula(Salary ~ ., data = hitters)
19 Variables  (and intercept)
           Forced in Forced out
AtBat          FALSE      FALSE
Hits           FALSE      FALSE
HmRun          FALSE      FALSE
Runs           FALSE      FALSE
RBI            FALSE      FALSE
Walks          FALSE      FALSE
Years          FALSE      FALSE
CAtBat         FALSE      FALSE
CHits          FALSE      FALSE
CHmRun         FALSE      FALSE
CRuns          FALSE      FALSE
CRBI           FALSE      FALSE
CWalks         FALSE      FALSE
LeagueN        FALSE      FALSE
DivisionW      FALSE      FALSE
PutOuts        FALSE      FALSE
Assists        FALSE      FALSE
Errors         FALSE      FALSE
NewLeagueN     FALSE      FALSE
1 subsets of each size up to 8
Selection Algorithm: exhaustive
         AtBat Hits HmRun Runs RBI Walks Years
1  ( 1 ) " "   " "  " "   " "  " " " " " "   " "
2  ( 1 ) " "   "*"  " "   " "  " " " " " "   " "
3  ( 1 ) " "   "*"  " "   " "  " " " " " "   " "
4  ( 1 ) " "   "*"  " "   " "  " " " " " "   " "
5  ( 1 ) "*"   "*"  " "   " "  " " " " " "   " "
6  ( 1 ) "*"   "*"  " "   " "  " " " " "*"   " "
7  ( 1 ) " "   "*"  " "   " "  " " " " "*"   " "
8  ( 1 ) "*"   "*"  " "   " "  " " " " "*"   " "
         CAtBat CHits CHmRun CRuns CRBI CWalks
1  ( 1 ) " "    " "   " "    " "   "*"  " "
2  ( 1 ) " "    " "   " "    " "   "*"  " "
3  ( 1 ) " "    " "   " "    " "   "*"  " "
4  ( 1 ) " "    " "   " "    " "   "*"  " "
5  ( 1 ) " "    " "   " "    " "   "*"  " "
6  ( 1 ) " "    " "   " "    " "   "*"  " "
7  ( 1 ) "*"    "*"   "*"    " "   " "  " "
8  ( 1 ) " "    " "   "*"    "*"   " "  "*"
         LeagueN DivisionW PutOuts Assists Errors
1  ( 1 ) " "     " "       " "     " "     " "
2  ( 1 ) " "     " "       " "     " "     " "
3  ( 1 ) " "     " "       "*"     " "     " "
4  ( 1 ) " "     "*"       "*"     " "     " "
5  ( 1 ) " "     "*"       "*"     " "     " "
6  ( 1 ) " "     "*"       "*"     " "     " "
7  ( 1 ) " "     "*"       "*"     " "     " "
8  ( 1 ) " "     "*"       "*"     " "     " "
         NewLeagueN
1  ( 1 ) " "
2  ( 1 ) " "
```

```
3  ( 1 ) " "
4  ( 1 ) " "
5  ( 1 ) " "
6  ( 1 ) " "
7  ( 1 ) " "
8  ( 1 ) " "
```

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```
model2<-regsubsets(Salary~.,data=hitters,nvmax=19)
summary(model2)
```

```
Subset selection object
Call: regsubsets.formula(Salary ~ ., data = hitters, nvmax = 19)
19 Variables  (and intercept)
           Forced in Forced out
AtBat          FALSE      FALSE
Hits           FALSE      FALSE
HmRun          FALSE      FALSE
Runs           FALSE      FALSE
RBI            FALSE      FALSE
Walks          FALSE      FALSE
Years          FALSE      FALSE
CAtBat         FALSE      FALSE
CHits          FALSE      FALSE
CHmRun         FALSE      FALSE
CRuns          FALSE      FALSE
CRBI           FALSE      FALSE
CWalks         FALSE      FALSE
LeagueN        FALSE      FALSE
DivisionW      FALSE      FALSE
PutOuts        FALSE      FALSE
Assists        FALSE      FALSE
Errors         FALSE      FALSE
NewLeagueN     FALSE      FALSE
1 subsets of each size up to 19
Selection Algorithm: exhaustive
          AtBat Hits HmRun Runs RBI Walks Years
1  ( 1 )  " "   " "  " "   " "  " " " "   " "
2  ( 1 )  " "   "*"  " "   " "  " " " "   " "
3  ( 1 )  " "   "*"  " "   " "  " " " "   " "
4  ( 1 )  " "   "*"  " "   " "  " " " "   " "
5  ( 1 )  "*"   "*"  " "   " "  " " " "   " "
6  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "
7  ( 1 )  " "   "*"  " "   " "  " " "*"   " "
8  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "
9  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "
10  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "
11  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "
12  ( 1 ) "*"   "*"  " "   "*"  " " "*"   " "
13  ( 1 ) "*"   "*"  " "   "*"  " " "*"   " "
14  ( 1 ) "*"   "*"  "*"   "*"  " " "*"   " "
15  ( 1 ) "*"   "*"  "*"   "*"  " " "*"   " "
16  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   " "
17  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   " "
18  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"
19  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"
          CAtBat CHits CHmRun CRuns CRBI CWalks
1  ( 1 )  " "    " "   " "    " "   "*"  " "
2  ( 1 )  " "    " "   " "    " "   "*"  " "
3  ( 1 )  " "    " "   " "    " "   "*"  " "
4  ( 1 )  " "    " "   " "    " "   "*"  " "
5  ( 1 )  " "    " "   " "    " "   "*"  " "
6  ( 1 )  " "    " "   " "    " "   "*"  " "
7  ( 1 )  "*"    "*"   "*"    " "   " "  " "
8  ( 1 )  " "    " "   "*"    "*"   " "  "*"
9  ( 1 )  "*"    " "   " "    "*"   "*"  "*"
```

```
10  ( 1 ) "*"    " "   " "    "*"   "*"  "*"
11  ( 1 ) "*"    " "   " "    "*"   "*"  "*"
12  ( 1 ) "*"    " "   " "    "*"   "*"  "*"
13  ( 1 ) "*"    " "   " "    "*"   "*"  "*"
14  ( 1 ) "*"    " "   " "    "*"   "*"  "*"
15  ( 1 ) "*"    "*"   " "    "*"   "*"  "*"
16  ( 1 ) "*"    "*"   " "    "*"   "*"  "*"
17  ( 1 ) "*"    "*"   " "    "*"   "*"  "*"
18  ( 1 ) "*"    "*"   " "    "*"   "*"  "*"
19  ( 1 ) "*"    "*"   "*"    "*"   "*"  "*"
          LeagueN DivisionW PutOuts Assists Errors
1   ( 1 )  " "     " "       " "     " "     " "
2   ( 1 )  " "     " "       " "     " "     " "
3   ( 1 )  " "     " "       "*"     " "     " "
4   ( 1 )  " "     "*"       "*"     " "     " "
5   ( 1 )  " "     "*"       "*"     " "     " "
6   ( 1 )  " "     "*"       "*"     " "     " "
7   ( 1 )  " "     "*"       "*"     " "     " "
8   ( 1 )  " "     "*"       "*"     " "     " "
9   ( 1 )  " "     "*"       "*"     " "     " "
10  ( 1 ) " "     "*"       "*"     "*"     " "
11  ( 1 ) "*"     "*"       "*"     "*"     " "
12  ( 1 ) "*"     "*"       "*"     "*"     " "
13  ( 1 ) "*"     "*"       "*"     "*"     "*"
14  ( 1 ) "*"     "*"       "*"     "*"     "*"
15  ( 1 ) "*"     "*"       "*"     "*"     "*"
16  ( 1 ) "*"     "*"       "*"     "*"     "*"
17  ( 1 ) "*"     "*"       "*"     "*"     "*"
18  ( 1 ) "*"     "*"       "*"     "*"     "*"
19  ( 1 ) "*"     "*"       "*"     "*"     "*"
          NewLeagueN
1   ( 1 )  " "
2   ( 1 )  " "
3   ( 1 )  " "
4   ( 1 )  " "
5   ( 1 )  " "
6   ( 1 )  " "
7   ( 1 )  " "
8   ( 1 )  " "
9   ( 1 )  " "
10  ( 1 ) " "
11  ( 1 ) " "
12  ( 1 ) " "
13  ( 1 ) " "
14  ( 1 ) " "
15  ( 1 ) " "
16  ( 1 ) " "
17  ( 1 ) "*"
18  ( 1 ) "*"
19  ( 1 ) "*"
```

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```
names(summary(model2))
```

```
[1] "which"  "rsq"    "rss"    "adjr2"  "cp"
[6] "bic"    "outmat" "obj"
```

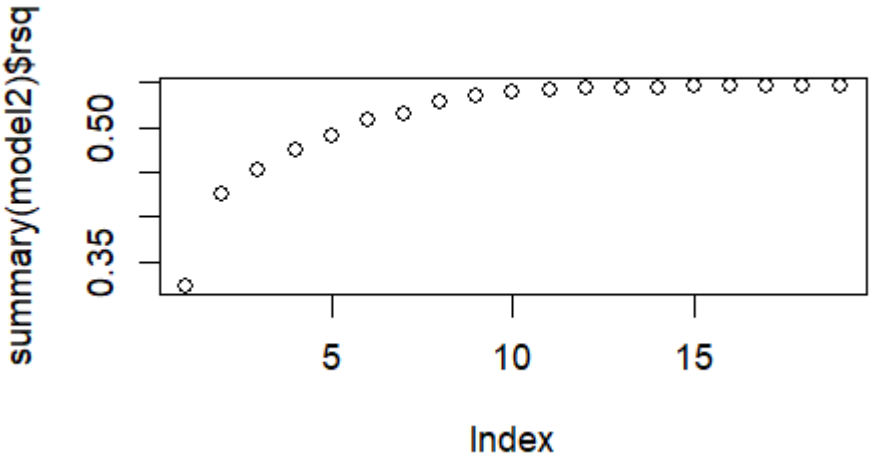
Hide

```
summary(model2)$rsq
```

```
 [1] 0.3214501 0.4252237 0.4514294 0.4754067
 [5] 0.4908036 0.5087146 0.5141227 0.5285569
 [9] 0.5346124 0.5404950 0.5426153 0.5436302
[13] 0.5444570 0.5452164 0.5454692 0.5457656
[17] 0.5459518 0.5460945 0.5461159
```

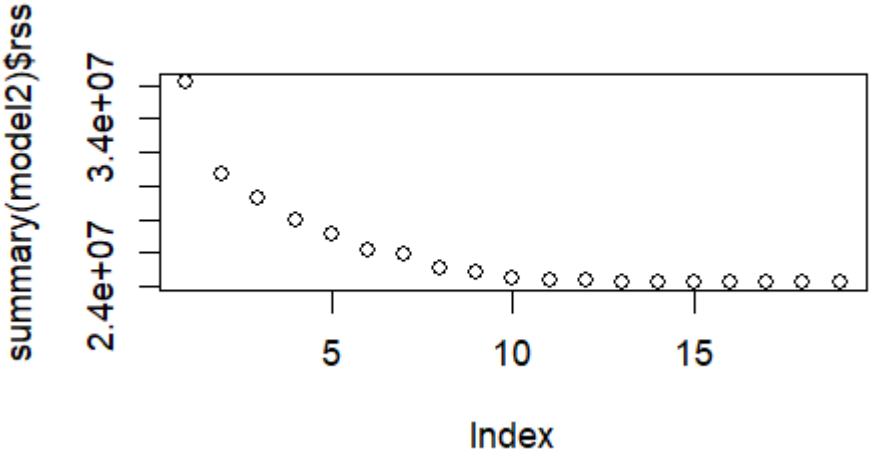
Hide

```
plot(summary(model2)$rsq)
```



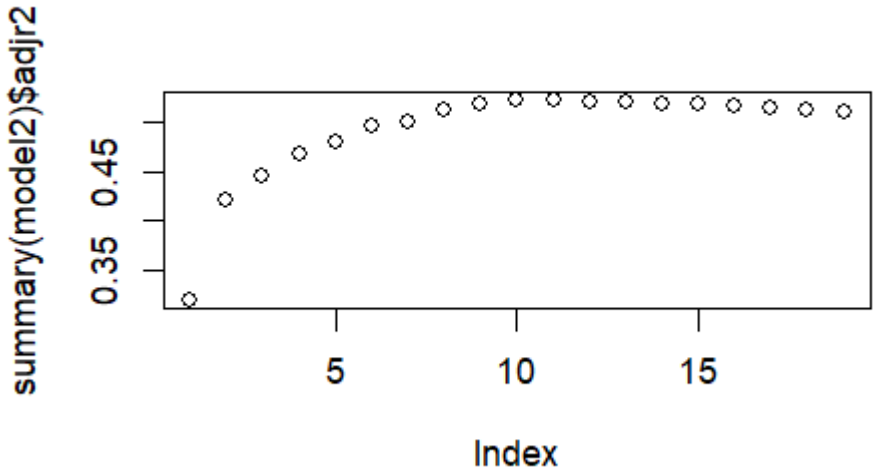
Hide

```
plot(summary(model2)$rss)
```



Hide

```
plot(summary(model2)$adjr2)
```



Hide

```
which.max(summary(model2)$adjr2)
```

```
[1] 11
```

Hide

```
coef(model2,11)
```

```
 (Intercept)        AtBat         Hits
 135.7512195   -2.1277482    6.9236994
       Walks       CAtBat        CRuns
   5.6202755   -0.1389914    1.4553310
        CRBI       CWalks      LeagueN
   0.7852528   -0.8228559   43.1116152
   DivisionW      PutOuts      Assists
-111.1460252    0.2894087    0.2688277
```

The figures indicate that R-squared increase as the number of variables in the subset increases and likewise the residual sum of squared (sum of squared errors) decreases as the size of the subsets increases. On the other hand the adjusted R-squared increases first and then decreases.

Forward stepwise selection: In this example, the best model identified by the forward stepwise selection is the same as that obtained by the best subset selection. It is also possible to run this algorithm using a backward method where you drop variables one a time rather add. In general, the solutions from these two methods can be different.

Hide

```
model3<-regsubsets(Salary~.,data=hitters,nvmax=19,method="forward")
which.max(summary(model3)$adjr2)
```

```
[1] 11
```

Hide

```
coef(model3,11)
```

```
 (Intercept)        AtBat         Hits
 135.7512195   -2.1277482    6.9236994
       Walks       CAtBat        CRuns
   5.6202755   -0.1389914    1.4553310
        CRBI       CWalks      LeagueN
   0.7852528   -0.8228559   43.1116152
   DivisionW      PutOuts      Assists
-111.1460252    0.2894087    0.2688277
```

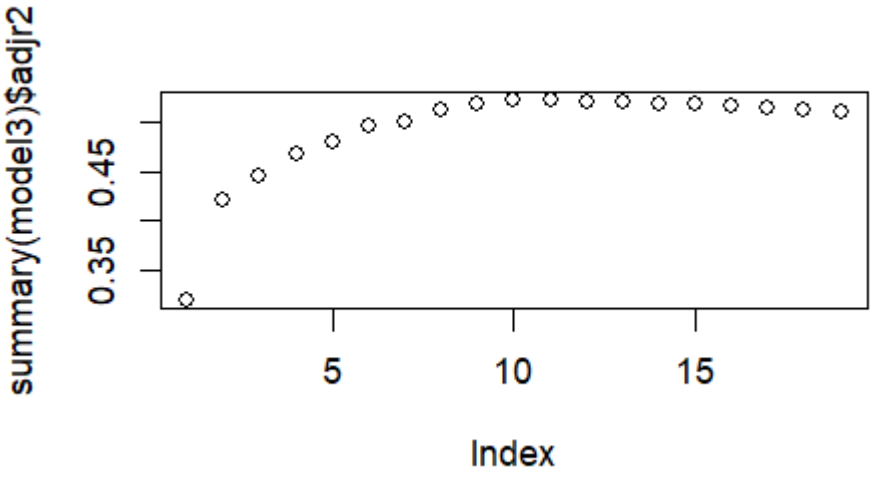
Hide

```
summary(model2)$adjr2-summary(model3)$adjr2
```

```
 [1] 3.330669e-16 1.110223e-16 0.000000e+00
 [4] 0.000000e+00 1.110223e-16 0.000000e+00
 [7] 9.185854e-04 4.314850e-04 1.110223e-16
[10] 1.110223e-16 1.110223e-16 0.000000e+00
[13] 0.000000e+00 2.220446e-16 1.110223e-16
[16] 0.000000e+00 0.000000e+00 0.000000e+00
[19] 0.000000e+00
```

Hide

```
plot(summary(model3)$adjr2)
```



Hide

```
model4<-regsubsets(Salary~.,data=hitters,nvmax=19,method="backward")
which.max(summary(model4)$adjr2)
```

```
[1] 11
```

Hide

```
coef(model3,11)
```

```
 (Intercept)        AtBat         Hits
 135.7512195   -2.1277482    6.9236994
       Walks       CAtBat        CRuns
   5.6202755   -0.1389914    1.4553310
        CRBI       CWalks      LeagueN
   0.7852528   -0.8228559   43.1116152
   DivisionW      PutOuts      Assists
-111.1460252    0.2894087    0.2688277
```

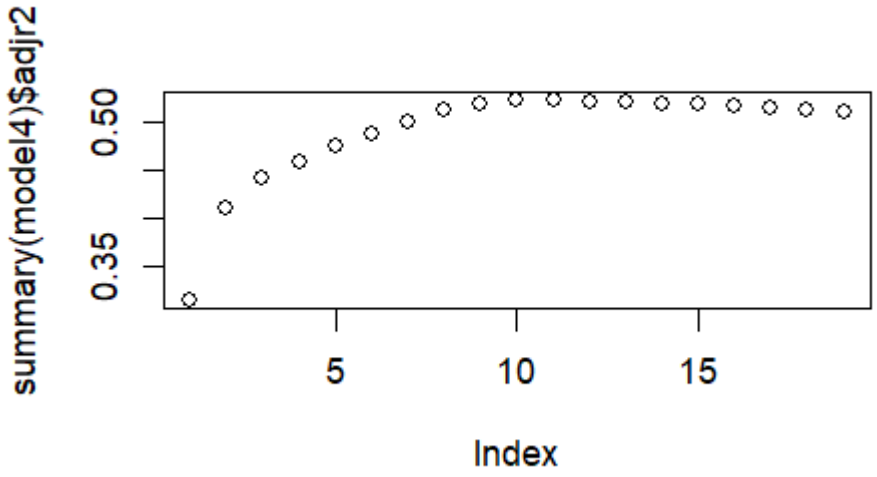
Hide

```
summary(model2)$adjr2-summary(model4)$adjr2
```

```
 [1]  4.862441e-03  1.052500e-02  2.997619e-03
 [4]  9.141119e-03  6.875981e-03  9.197808e-03
 [7]  5.191286e-04  4.314850e-04  1.110223e-16
[10]  2.220446e-16  2.220446e-16  0.000000e+00
[13] -1.110223e-16  2.220446e-16  1.110223e-16
[16]  0.000000e+00  0.000000e+00  0.000000e+00
[19]  0.000000e+00
```

Hide

```
plot(summary(model4)$adjr2)
```



The following is AI generated code to help with making the plots fancier.

Hide

```
par(mfrow = c(1, 3))

plot(summary(model2)$adjr2, type = "b", xlab = "# Variables", ylab = "Adj. R²",
     main = "Exhaustive", pch = 20)
abline(v = which.max(summary(model2)$adjr2), col = "red", lty = 2)
```

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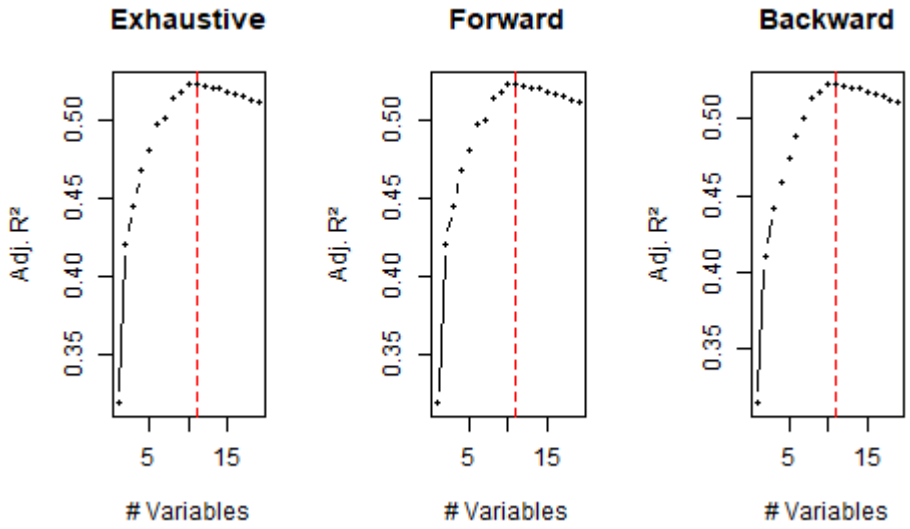
```
plot(summary(model3)$adjr2, type = "b", xlab = "# Variables", ylab = "Adj. R²",
     main = "Forward", pch = 20)
abline(v = which.max(summary(model3)$adjr2), col = "red", lty = 2)
```

Hide

```
plot(summary(model4)$adjr2, type = "b", xlab = "# Variables", ylab = "Adj. R²",
     main = "Backward", pch = 20)
abline(v = which.max(summary(model4)$adjr2), col = "red", lty = 2)
```

Hide

```
par(mfrow = c(1, 1))
```



LASSO

The generalized linear model with penalized maximum likelihood package glmnet in R implements the LASSO method. To run the glmnet() function, we need to pass in the arguments as X (input matrix), y (output vector), rather than the y~X format that we used thus far. The model.matrix() function produces a matrix corresponding to the 19 predictors and the intercept and helps transform qualitative variables into dummy quantitative variables. This is important since glmnet() works only with quantitative variables. We will work with the X matrix directly from this point onwards and get the solutions. We will also drop the first column corresponding to the intercept term that is introduced in X since glmnet() by default estimates the intercept and outputs it.

Hide

```
#install.packages("glmnet")
library(glmnet)
X <- model.matrix(Salary ~ ., data = hitters)[, -1]
# drop intercept column
str(X)
```

```
 num [1:263, 1:19] 315 479 496 321 594 185 298 323 401 574 ...
 - attr(*, "dimnames")=List of 2
  ..$ : chr [1:263] "2" "3" "4" "5" ...
  ..$ : chr [1:19] "AtBat" "Hits" "HmRun" "Runs" ...
```

Hide

```
colnames(X)
```

```
 [1] "AtBat"      "Hits"       "HmRun"
 [4] "Runs"       "RBI"        "Walks"
 [7] "Years"      "CAtBat"     "CHits"
[10] "CHmRun"     "CRuns"      "CRBI"
[13] "CWalks"     "LeagueN"    "DivisionW"
[16] "PutOuts"    "Assists"    "Errors"
[19] "NewLeagueN"
```

Hide

```
y <- hitters$Salary
str(X)
```

```
 num [1:263, 1:19] 315 479 496 321 594 185 298 323 401 574 ...
 - attr(*, "dimnames")=List of 2
  ..$ : chr [1:263] "2" "3" "4" "5" ...
  ..$ : chr [1:19] "AtBat" "Hits" "HmRun" "Runs" ...
```

We now choose a range for the lambda parameters and create a training and test set. We then build the LASSO model on this data. The output from the model provides the Df (non of nonzeros), %Dev and Lambda values. The deviance measure is given as 2(loglike_sat - loglike), where loglike_sat is the log-likelihood for the saturated model (a model with a free parameter per observation) and loglike is the log-likelihood of your model. Smaller the deviance, better the fit to the data. The deviance measure for linear regression is exactly the sum of squared errors. Null deviance is defined to be 2(loglike_sat - loglike(NULL)) where the NULL model refers to the intercept model only. The null deviance for linear regression is exactly total sum of squares. The deviance ratio is dev.ratio=1-deviance/nulldev. You can interpret the dev.ratio simply as the Rsquared value for regression problems (this will be sufficient for our analysis). Higher the deviance ratio, better the fit to the data. As lambda decreases, the dev.ratio increases (more importance given to model fit than model complexity). The a0 output gives the intercept estimate and beta gives the beta estimates. We run glmnet here with the default lambda values. We can specify them too if we want.

Hide

```
set.seed(1)
train <- sample(1:nrow(X),nrow(X)/2)
test <- -train
modellasso <- glmnet(X[train,],y[train])
summary(modellasso)
```

```
          Length Class     Mode
a0         100   -none-    numeric
beta      1900   dgCMatrix S4
df         100   -none-    numeric
dim          2   -none-    numeric
lambda     100   -none-    numeric
dev.ratio  100   -none-    numeric
nulldev      1   -none-    numeric
npasses      1   -none-    numeric
jerr         1   -none-    numeric
offset       1   -none-    logical
call         3   -none-    call
nobs         1   -none-    numeric
```

Hide

```
modellasso
```

```
Call:  glmnet(x = X[train, ], y = y[train])

    Df  %Dev  Lambda
1    0  0.00 264.500
2    1  6.57 241.000
3    1 12.03 219.600
4    2 16.64 200.100
5    2 20.51 182.300
6    3 24.00 166.100
7    3 27.88 151.400
8    3 31.11 137.900
9    3 33.78 125.700
10   3 36.00 114.500
11   3 37.85 104.300
12   3 39.38  95.050
13   4 40.72  86.610
14   4 42.28  78.920
15   6 43.60  71.910
16   6 44.73  65.520
17   6 45.67  59.700
18   7 46.55  54.390
19   8 47.33  49.560
20   8 48.00  45.160
21   8 48.55  41.150
22  10 49.02  37.490
23   8 49.43  34.160
24   8 49.78  31.130
25   9 50.14  28.360
26   9 50.47  25.840
27  10 50.81  23.550
28  10 51.10  21.450
29  10 51.35  19.550
30  10 51.55  17.810
31  10 51.72  16.230
32  11 51.95  14.790
33  11 52.36  13.470
34  11 52.70  12.280
35  11 52.99  11.190
36  10 53.21  10.190
37  10 53.38   9.287
38  10 53.52   8.462
39  10 53.63   7.710
40  11 53.74   7.025
41  11 53.87   6.401
42  12 54.09   5.832
43  12 54.28   5.314
44  12 54.44   4.842
45  12 54.57   4.412
46  13 54.73   4.020
47  13 54.91   3.663
48  13 55.06   3.338
49  13 55.19   3.041
50  13 55.29   2.771
51  14 55.44   2.525
```

```
52  14 55.87   2.300
53  14 56.25   2.096
54  15 56.57   1.910
55  15 56.86   1.740
56  15 57.10   1.586
57  15 57.30   1.445
58  15 57.47   1.316
59  16 57.61   1.199
60  17 57.73   1.093
61  17 57.85   0.996
62  17 57.96   0.907
63  17 58.05   0.827
64  18 58.14   0.753
65  17 58.20   0.686
66  17 58.25   0.625
67  16 58.29   0.570
68  16 58.32   0.519
69  16 58.34   0.473
70  16 58.36   0.431
71  16 58.38   0.393
72  17 58.40   0.358
73  17 58.41   0.326
74  17 58.42   0.297
75  19 58.44   0.271
76  19 58.45   0.247
77  19 58.49   0.225
78  19 58.49   0.205
79  19 58.52   0.187
80  19 58.53   0.170
81  19 58.54   0.155
82  19 58.55   0.141
83  19 58.56   0.129
84  19 58.57   0.117
85  19 58.58   0.107
86  19 58.58   0.097
87  19 58.59   0.089
88  19 58.59   0.081
89  19 58.59   0.074
90  19 58.60   0.067
91  19 58.60   0.061
92  19 58.60   0.056
93  19 58.60   0.051
94  19 58.60   0.046
95  19 58.61   0.042
96  19 58.61   0.038
97  19 58.61   0.035
98  19 58.61   0.032
99  19 58.61   0.029
100 19 58.61   0.026
```

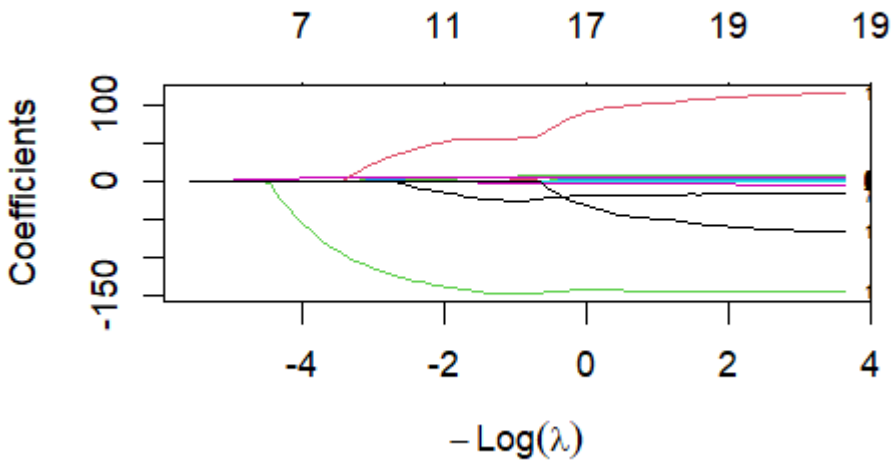
Hide

```
deviance(modellasso)
```

```
  [1] 23667021 22111050 20819255 19729261 18812379
  [6] 17987594 17067936 16305047 15671748 15145887
 [11] 14709390 14346937 14030529 13660055 13349004
 [16] 13080381 12857190 12648873 12465519 12307920
 [21] 12176866 12065305 11967669 11886379 11799332
 [26] 11721223 11642503 11573553 11514204 11466000
 [31] 11425790 11371598 11275818 11193426 11124830
 [36] 11074639 11034246 11000792 10973373 10948995
 [41] 10918261 10865633 10820567 10783366 10752548
 [46] 10714120 10671010 10635053 10605317 10580452
 [51] 10546091 10443113 10353636 10277597 10209628
 [56] 10153039 10105839 10066378 10032826 10004488
 [61]  9976542  9949474  9927139  9907822  9891754
 [66]  9881002  9872464  9865121  9859088  9853913
 [71]  9849692  9846101  9842712  9840754  9836985
 [76]  9833276  9824988  9823291  9816430  9815432
 [81]  9812125  9809295  9807128  9805331  9803845
 [86]  9802571  9801484  9800562  9799763  9799074
 [91]  9798466  9797966  9797527  9797128  9796796
 [96]  9796509  9796249  9796028  9795842  9795662
```

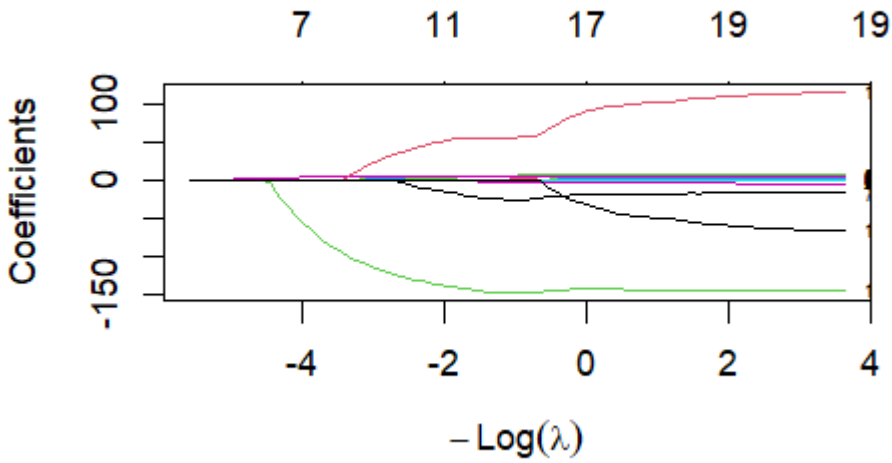
Hide

```
plot(modellasso,label=TRUE)
```



Hide

```
plot(modellasso,xvar="lambda",label=TRUE)
```



We see from the plot that as lambda increases, many of the coefficients get close to zero. We can retrieve these coefficients at specific values of lambda. The a0 term keeps track of the intercept in glmnet. Note that the number of non-zero coefficients does not change in a fully monotonic way, as lambda increases or decreases but the general trend is that as lambda increases, the number of zero coefficients increases (or number of non-zero coefficients decrease).

Hide

```
modellasso$df
```

```
  [1]  0  1  1  2  2  3  3  3  3  3  3  3  4  4  6
 [16]  6  6  7  8  8  8 10  8  8  9  9 10 10 10 10
 [31] 10 11 11 11 11 10 10 10 10 11 11 12 12 12 12
 [46] 13 13 13 13 13 14 14 14 15 15 15 15 15 16 17
 [61] 17 17 17 18 17 17 16 16 16 16 16 17 17 17 19
 [76] 19 19 19 19 19 19 19 19 19 19 19 19 19 19 19
 [91] 19 19 19 19 19 19 19 19 19 19
```

Hide

```
modellasso$a0
```

```
       s0        s1        s2        s3        s4
531.86537 505.49838 481.47375 460.51767 442.09143
       s5        s6        s7        s8        s9
421.58253 389.38027 359.92818 333.07966 308.63793
      s10       s11       s12       s13       s14
286.34755 266.05749 248.41927 239.17924 230.00988
      s15       s16       s17       s18       s19
219.78200 210.38800 200.74720 190.94915 181.34000
      s20       s21       s22       s23       s24
172.72411 164.64615 157.17910 150.17758 140.73484
      s25       s26       s27       s28       s29
131.16210 122.46907 114.90372 107.80904 101.40317
      s30       s31       s32       s33       s34
 95.54681  93.53147 100.64108 107.72647 114.28785
      s35       s36       s37       s38       s39
117.62780 119.84177 121.90781 123.69211 125.72654
      s40       s41       s42       s43       s44
129.28822 140.89645 151.71947 161.52186 170.42527
      s45       s46       s47       s48       s49
179.89838 189.90348 198.97075 207.19205 214.71282
      s50       s51       s52       s53       s54
222.11509 225.18344 227.97429 230.39376 232.80785
      s55       s56       s57       s58       s59
234.98599 236.97476 238.72409 240.73568 242.04725
      s60       s61       s62       s63       s64
243.48182 246.08704 248.51711 250.79599 253.43746
      s65       s66       s67       s68       s69
256.16948 258.59577 260.59389 262.58883 264.23261
      s70       s71       s72       s73       s74
265.81974 267.26457 268.60895 269.52522 270.63066
      s75       s76       s77       s78       s79
271.37281 270.94552 271.55508 271.42364 271.85351
      s80       s81       s82       s83       s84
272.69812 273.02675 273.21919 273.37165 273.50387
      s85       s86       s87       s88       s89
273.62310 273.73075 273.82844 273.91767 273.99861
      s90       s91       s92       s93       s94
274.07226 274.13785 274.19868 274.25473 274.30409
      s95       s96       s97       s98       s99
274.34926 274.39121 274.42860 274.46225 274.49476
```

Hide

```
as.matrix(modellasso$beta)
```

```
           s0         s1        s2        s3
AtBat       0 0.00000000 0.0000000 0.0000000
Hits        0 0.00000000 0.0000000 0.0000000
HmRun       0 0.00000000 0.0000000 0.0000000
Runs        0 0.00000000 0.0000000 0.0000000
RBI         0 0.00000000 0.0000000 0.0000000
Walks       0 0.00000000 0.0000000 0.0000000
Years       0 0.00000000 0.0000000 0.0000000
CAtBat      0 0.00000000 0.0000000 0.0000000
CHits       0 0.00000000 0.0000000 0.0000000
CHmRun      0 0.00000000 0.0000000 0.0877153
                  s4        s5        s6        s7
AtBat      0.0000000 0.0000000 0.0000000 0.0000000
Hits       0.0000000 0.0000000 0.0000000 0.0000000
HmRun      0.0000000 0.0000000 0.0000000 0.0000000
Runs       0.0000000 0.0000000 0.0000000 0.0000000
RBI        0.0000000 0.0000000 0.0000000 0.0000000
Walks      0.0000000 0.1095612 0.6210547 1.0880248
Years      0.0000000 0.0000000 0.0000000 0.0000000
CAtBat     0.0000000 0.0000000 0.0000000 0.0000000
CHits      0.0000000 0.0000000 0.0000000 0.0000000
CHmRun     0.2303683 0.3429256 0.4232717 0.4888717
                  s8        s9       s10       s11
AtBat      0.0000000 0.0000000 0.0000000 0.0000000
Hits       0.0000000 0.0000000 0.0000000 0.0000000
HmRun      0.0000000 0.0000000 0.0000000 0.0000000
Runs       0.0000000 0.0000000 0.0000000 0.0000000
RBI        0.0000000 0.0000000 0.0000000 0.0000000
Walks      1.5136197 1.9012226 2.2545612 2.5763397
Years      0.0000000 0.0000000 0.0000000 0.0000000
CAtBat     0.0000000 0.0000000 0.0000000 0.0000000
CHits      0.0000000 0.0000000 0.0000000 0.0000000
CHmRun     0.5477657 0.6029035 0.6517801 0.6976866
                 s12         s13           s14
AtBat       0.0000000   0.0000000   0.000000000
Hits        0.0000000   0.0000000   0.000000000
HmRun       0.0000000   0.0000000   0.000000000
Runs        0.0000000   0.0000000   0.007804832
RBI         0.0000000   0.0000000   0.023898403
Walks       2.8723279   3.1556696   3.393837658
Years       0.0000000   0.0000000   0.000000000
CAtBat      0.0000000   0.0000000   0.000000000
CHits       0.0000000   0.0000000   0.000000000
CHmRun      0.7317752   0.7632056   0.778216900
                 s15         s16           s17
AtBat        0.00000000   0.0000000   0.000000000
Hits         0.00000000   0.0000000   0.000000000
HmRun        0.00000000   0.0000000   0.000000000
Runs         0.08093599   0.1516809   0.214117019
RBI          0.07270051   0.1151076   0.141388496
Walks        3.54052128   3.6724799   3.758336099
Years        0.00000000   0.0000000   0.000000000
CAtBat       0.00000000   0.0000000   0.000000000
CHits        0.00000000   0.0000000   0.000000000
CHmRun       0.78330289   0.7858554   0.780583983
```

```
                  s18         s19          s20
AtBat        0.00000000   0.0000000   0.00000000
Hits         0.01498891   0.1186950   0.21745394
HmRun        0.00000000   0.0000000   0.00000000
Runs         0.26661387   0.1786221   0.09732585
RBI          0.15083828   0.1236041   0.09019558
Walks        3.81326488   3.8775423   3.93628016
Years        0.00000000   0.0000000   0.00000000
CAtBat       0.00000000   0.0000000   0.00000000
CHits        0.00000000   0.0000000   0.00000000
CHmRun       0.76592306   0.7944974   0.83427366
                  s21          s22          s23
AtBat        0.000000000  0.00000000   0.00000000
Hits         0.295875262  0.33910800   0.34967540
HmRun        0.064053026  0.39035097   0.56712264
Runs         0.012301833  0.00000000   0.00000000
RBI          0.072512331  0.00000000   0.00000000
Walks        3.998237124  4.04263883   4.08292259
Years        0.000000000  0.00000000   0.00000000
CAtBat       0.000000000  0.00000000   0.00000000
CHits        0.005243442  0.01252331   0.02693188
CHmRun       0.871725632  0.87222074   0.91414101
                  s24          s25
AtBat         0.00000000   0.00000000
Hits          0.37995494   0.41376625
HmRun         0.72709544   0.87818639
Runs          0.00000000   0.00000000
RBI           0.00000000   0.00000000
Walks         4.08592669   4.07837663
Years         0.00000000   0.00000000
CAtBat        0.00000000   0.00000000
CHits         0.03973360   0.04971208
CHmRun        0.96019641   0.99290548
                  s26          s27
AtBat         0.00000000   0.00000000
Hits          0.41171670   0.39018482
HmRun         1.11310560   1.39778961
Runs          0.00000000   0.00000000
RBI           0.00000000   0.00000000
Walks         4.07631819   4.08465641
Years         0.00000000   0.00000000
CAtBat        0.00000000   0.00000000
CHits         0.06032560   0.06582110
CHmRun        1.04081759   1.04924490
                  s28          s29
AtBat         0.00000000   0.00000000
Hits          0.36119995   0.34303485
HmRun         1.70277323   1.94107401
Runs          0.00000000   0.00000000
RBI           0.00000000   0.00000000
Walks         4.09744804   4.10348570
Years         0.00000000   0.00000000
CAtBat        0.00000000   0.00000000
CHits         0.07683496   0.08378951
CHmRun        1.07732680   1.09844269
                  s30          s31
```

```
AtBat         0.00000000    0.00000000
Hits          0.32556062    0.29143832
HmRun         2.16254406    2.37477111
Runs          0.00000000    0.00000000
RBI           0.00000000    0.00000000
Walks         4.10949673    4.11106813
Years         0.00000000   -0.76442030
CAtBat        0.00000000    0.00000000
CHits         0.09073734    0.10450327
CHmRun        1.11988415    1.14006395
                    s32           s33           s34
AtBat         0.0000000    0.0000000    0.00000000
Hits          0.2029085    0.1119611    0.02659006
HmRun         2.6600735    2.9678217    3.26481054
Runs          0.0000000    0.0000000    0.00000000
RBI           0.0000000    0.0000000    0.00000000
Walks         4.1099380    4.1136122    4.11875476
Years        -3.4505573   -6.0200626   -8.37700568
CAtBat        0.0000000    0.0000000    0.00000000
CHits         0.1309559    0.1550448    0.17600665
CHmRun        1.0987462    1.0351605    0.96360902
                    s35           s36           s37
AtBat         0.0000000    0.0000000    0.0000000
Hits          0.0000000    0.0000000    0.0000000
HmRun         3.3875312    3.4239422    3.4527540
Runs          0.0000000    0.0000000    0.0000000
RBI           0.0000000    0.0000000    0.0000000
Walks         4.0919122    4.0487555    4.0095547
Years       -10.2055368  -11.8252122  -13.2996313
CAtBat        0.0000000    0.0000000    0.0000000
CHits         0.1918772    0.2052751    0.2172280
CHmRun        0.9298302    0.9106627    0.8938979
                    s38           s39           s40
AtBat         0.0000000    0.00000000    0.0000000
Hits          0.0000000    0.00000000    0.0000000
HmRun         3.4805209    3.50583637    3.5819252
Runs          0.0000000    0.00000000    0.0000000
RBI           0.0000000    0.00000000    0.0000000
Walks         3.9747491    3.94331284    3.9172989
Years       -14.6169107  -15.86828408  -17.0263001
CAtBat        0.0000000    0.00000000    0.0000000
CHits         0.2278926    0.23764801    0.2460677
CHmRun        0.8789078    0.86167343    0.8429630
                    s41           s42
AtBat        -0.04389081   -0.08706944
Hits          0.00000000    0.00000000
HmRun         4.02705026    4.47870279
Runs          0.00000000    0.00000000
RBI           0.00000000    0.00000000
Walks         3.98899742    4.06333631
Years       -18.72567091  -20.25674145
CAtBat        0.00000000    0.00000000
CHits         0.25858285    0.26968276
CHmRun        0.75620052    0.66500649
                    s43           s44           s45
AtBat        -0.1261800   -0.1616992   -0.2357964
```

```
Hits          0.0000000    0.0000000    0.1503145
HmRun         4.8880704    5.2595941    5.5215808
Runs          0.0000000    0.0000000    0.0000000
RBI           0.0000000    0.0000000    0.0000000
Walks         4.1309976    4.1925304    4.2394764
Years       -21.6387651  -22.8928771  -23.9331886
CAtBat        0.0000000    0.0000000    0.0000000
CHits         0.2794007    0.2881900    0.2931407
CHmRun        0.5809432    0.5047707    0.4581748
                    s46          s47          s48
AtBat        -0.3430320   -0.4417090   -0.5310141
Hits          0.4207553    0.6717305    0.8988989
HmRun         5.7201031    5.8932897    6.0488698
Runs          0.0000000    0.0000000    0.0000000
RBI           0.0000000    0.0000000    0.0000000
Walks         4.2824817    4.3211871    4.3560846
Years       -24.7157142  -25.4057224  -26.0347130
CAtBat        0.0000000    0.0000000    0.0000000
CHits         0.2937731    0.2942690    0.2949078
CHmRun        0.4350495    0.4185656    0.4051273
                    s49           s50
AtBat        -0.6130936 -6.974205e-01
Hits          1.1080601  1.316676e+00
HmRun         6.1905047  6.365296e+00
Runs          0.0000000  0.000000e+00
RBI           0.0000000  0.000000e+00
Walks         4.3880229  4.422774e+00
Years       -26.6049110 -2.688074e+01
CAtBat        0.0000000 -5.529584e-03
CHits         0.2954473  3.117413e-01
CHmRun        0.3933633  3.757822e-01
                    s51           s52
AtBat        -0.68676536   -0.67328543
Hits          1.18967443    1.06104880
HmRun         6.57606149    6.75337728
Runs          0.00000000    0.00000000
RBI           0.00000000    0.00000000
Walks         4.42752733    4.42820350
Years       -25.75316318  -24.69553852
CAtBat       -0.05384494   -0.09998821
CHits         0.48643509    0.65507480
CHmRun        0.50133718    0.63765618
                    s53          s54          s55
AtBat        -0.6586566   -0.6419337   -0.6265019
Hits          0.9377095    0.8158301    0.7043822
HmRun         6.9073613    7.0290653    7.1393125
Runs          0.0000000    0.0000000    0.0000000
RBI           0.0000000    0.0000000    0.0000000
Walks         4.4297979    4.4371295    4.4436889
Years       -23.6881389  -22.7103271  -21.8180747
CAtBat       -0.1428055   -0.1836075   -0.2207736
CHits         0.8116446    0.9610909    1.0972308
CHmRun        0.7672472    0.8951414    1.0119011
                    s56          s57           s58
AtBat        -0.6124100   -0.5990622 -5.870480e-01
Hits          0.6026829    0.5087318  4.269444e-01
```

```
HmRun         7.2397771    7.3302901  7.431346e+00
Runs          0.0000000    0.0000000  0.000000e+00
RBI           0.0000000    0.0000000  0.000000e+00
Walks         4.4496497    4.4548539  4.435141e+00
Years       -21.0053472  -20.2529359 -1.962572e+01
CAtBat       -0.2546619   -0.2856818 -3.144890e-01
CHits         1.2213718    1.3350223  1.438727e+00
CHmRun        1.1183617    1.2166799  1.298378e+00
                    s59          s60
AtBat      -5.673185e-01   -0.54759522
Hits        3.289713e-01    0.22964936
HmRun       7.587644e+00    7.69580062
Runs        0.000000e+00    0.00000000
RBI         0.000000e+00    0.00000000
Walks       4.389496e+00    4.36436918
Years      -1.903680e+01  -18.46130111
CAtBat     -3.419655e-01   -0.36929810
CHits       1.534972e+00    1.63976579
CHmRun      1.345523e+00    1.42645912
                    s61          s62
AtBat        -0.53281982   -0.52039249
Hits          0.15194968    0.08504343
HmRun         7.78795214    7.86907401
Runs          0.00000000    0.00000000
RBI           0.00000000    0.00000000
Walks         4.34571978    4.32937912
Years       -18.25488089  -18.08592383
CAtBat       -0.39142736   -0.41106188
CHits         1.74866333    1.84654525
CHmRun        1.54409570    1.65236207
                    s63          s64
AtBat        -0.50955982   -0.51708614
Hits          0.01009352    0.00000000
HmRun         7.84880085    7.77757656
Runs          0.00000000    0.00000000
RBI           0.05585887    0.11674142
Walks         4.31015356    4.29501834
Years       -17.93596458  -17.79235548
CAtBat       -0.42936081   -0.44454064
CHits         1.94024494    2.02028507
CHmRun        1.76392428    1.87967472
                    s65          s66          s67
AtBat      -5.245988e-01   -0.5301626   -0.5348956
Hits        0.000000e+00    0.0000000    0.0000000
HmRun       7.814991e+00    7.9021838    7.9862144
Runs        0.000000e+00    0.0000000    0.0000000
RBI         1.324328e-01    0.1258950    0.1182084
Walks       4.283094e+00    4.2722950    4.2636457
Years      -1.780288e+01  -17.8161205  -17.7739683
CAtBat     -4.556036e-01   -0.4656498   -0.4751680
CHits       2.073094e+00    2.1168830    2.1579801
CHmRun      1.917379e+00    1.9267525    1.9359981
                    s68          s69          s70
AtBat        -0.5395601   -0.5434582   -0.5471831
Hits          0.0000000    0.0000000    0.0000000
HmRun         8.0589775    8.1292599    8.1907441
```

```
Runs          0.0000000    0.0000000    0.0000000
RBI           0.1124021    0.1057942    0.1008876
Walks         4.2551901    4.2480742    4.2412977
Years       -17.7759689  -17.7360789  -17.7271498
CAtBat       -0.4836757   -0.4916314   -0.4986073
CHits         2.1947560    2.2290690    2.2594441
CHmRun        1.9440216    1.9518105    1.9583886
                    s71          s72
AtBat     -5.506137e-01   -0.55680609
Hits       0.000000e+00    0.00000000
HmRun      8.247522e+00    8.28176237
Runs       9.917932e-04    0.03393223
RBI        9.549709e-02    0.08546903
Walks      4.234870e+00    4.22289714
Years     -1.771514e+01  -17.67612673
CAtBat    -5.050723e-01   -0.51126343
CHits      2.287517e+00    2.31540914
CHmRun     1.964632e+00    1.97485774
                    s73          s74
AtBat        -0.56070749 -5.583130e-01
Hits          0.00000000 -4.901684e-02
HmRun         8.31803987  8.328538e+00
Runs          0.07350006  1.300860e-01
RBI           0.06876119  7.488906e-02
Walks         4.20802316  4.192923e+00
Years       -17.73754880 -1.765420e+01
CAtBat       -0.51345109 -5.219481e-01
CHits         2.33127369  2.368084e+00
CHmRun        1.98069080  1.992236e+00
                    s75          s76
AtBat        -0.55057822   -0.52492002
Hits         -0.13611806   -0.29358911
HmRun         8.26403646    7.98605756
Runs          0.25752804    0.34054415
RBI           0.09675888    0.22265491
Walks         4.16173746    4.12308487
Years       -17.77094524  -17.42023461
CAtBat       -0.52516454   -0.54055157
CHits         2.40161863    2.48664582
CHmRun        2.02478360    2.18820733
                    s77          s78          s79
AtBat        -0.52229530   -0.5001954   -0.4984410
Hits         -0.33818021   -0.4880850   -0.5190280
HmRun         7.95933849    7.6705790    7.6607036
Runs          0.41783578    0.5123787    0.5698516
RBI           0.23130401    0.3574294    0.3586530
Walks         4.10440608    4.0660645    4.0529725
Years       -17.53179529  -17.2711701  -17.3510711
CAtBat       -0.54155047   -0.5550739   -0.5556771
CHits         2.50370059    2.5844711    2.5960810
CHmRun        2.20448004    2.3649912    2.3740926
                   s80          s81          s82
AtBat        -0.4916241   -0.4832827   -0.4753997
Hits         -0.6082420   -0.6855206   -0.7507864
HmRun         7.4946583    7.3442602    7.2295659
Runs          0.6836980    0.7602721    0.8198393
```

```
RBI          0.4241907    0.4849315    0.5328688
Walks        4.0214489    3.9986218    3.9795885
Years      -17.4575999  -17.4306900  -17.3867222
CAtBat      -0.5597045   -0.5649536   -0.5697249
CHits        2.6382255    2.6781677    2.7115317
CHmRun       2.4523323    2.5315217    2.5944710
                   s83          s84          s85
AtBat       -0.4680592   -0.4614609   -0.4553907
Hits        -0.8095269   -0.8618238   -0.9097431
HmRun        7.1308939    7.0448661    6.9664515
Runs         0.8722186    0.9185587    0.9609293
RBI          0.5746379    0.6113412    0.6448167
Walks        3.9624250    3.9470957    3.9330420
Years      -17.3428070  -17.3030856  -17.2663992
CAtBat      -0.5740918   -0.5779910   -0.5815691
CHits        2.7413250    2.7677329    2.7919041
CHmRun       2.6493736    2.6975238    2.7414602
                   s86          s87          s88
AtBat       -0.4498613   -0.4448708   -0.4402837
Hits        -0.9533190   -0.9926569   -1.0287913
HmRun        6.8953503    6.8313210    6.7724440
Runs         0.9994110    1.0341538    1.0660612
RBI          0.6751997    0.7026057    0.7277745
Walks        3.9202602    3.9087133    3.8981111
Years      -17.2328690  -17.2026976  -17.1749055
CAtBat      -0.5848256   -0.5877638   -0.5904641
CHits        2.8138731    2.8336927    2.8519034
CHmRun       2.7813344    2.8172701    2.8502985
                   s89          s90          s91
AtBat       -0.4360972   -0.4321793   -0.4287865
Hits        -1.0617493   -1.0924995   -1.1192052
HmRun        6.7187427    6.6684885    6.6251740
Runs         1.0951459    1.1222285    1.1457657
RBI          0.7507277    0.7721351    0.7907181
Walks        3.8884455    3.8794443    3.8716135
Years      -17.1494366  -17.1252853  -17.1045126
CAtBat      -0.5929287   -0.5952344   -0.5972330
CHits        2.8685159    2.8840331    2.8974856
CHmRun       2.8804291    2.9086002    2.9329645
                   s92          s93          s94
AtBat       -0.4256533   -0.4226520   -0.4200395
Hits        -1.1438749   -1.1674123   -1.1879332
HmRun        6.5850806    6.5465658    6.5131965
Runs         1.1675369    1.1882677    1.2063333
RBI          0.8078826    0.8242609    0.8385327
Walks        3.8643751    3.8574869    3.8514828
Years      -17.0854824  -17.0669776  -17.0507848
CAtBat      -0.5990772   -0.6008425   -0.6023816
CHits        2.9099134    2.9217939    2.9321419
CHmRun       2.9554836    2.9770585    2.9958214
                   s95          s96          s97
AtBat       -0.4176825   -0.4154527   -0.4134777
Hits        -1.2064699   -1.2239677   -1.2394826
HmRun        6.4831026    6.4545554    6.4293285
Runs         1.2226686    1.2380793    1.2517457
RBI          0.8514226    0.8635921    0.8743754
```

```
Walks         3.8460524    3.8409297    3.8363917
Years       -17.0363031  -17.0225701  -17.0103557
CAtBat       -0.6037697   -0.6050816   -0.6062447
CHits         2.9414831    2.9503098    2.9581331
CHmRun        3.0127471    3.0287610    3.0429471
                    s98          s99
AtBat        -0.4117550   -0.4100302
Hits         -1.2530631   -1.2665960
HmRun         6.4073664    6.3852223
Runs          1.2637304    1.2756560
RBI           0.8838123    0.8932321
Walks         3.8324146    3.8284479
Years       -16.9998147  -16.9892716
CAtBat       -0.6072602   -0.6082740
CHits         2.9649712    2.9717990
CHmRun        3.0553285    3.0677219
 [ reached 'max' / getOption("max.print") -- omitted 9 rows ]
```

Hide

```
modellasso$lambda[1]
```

```
[1] 264.4958
```

Hide

```
modellasso$a0[1]
```

```
      s0
531.8654
```

Hide

```
modellasso$beta[,1]
```

```
     AtBat       Hits      HmRun       Runs
         0          0          0          0
       RBI      Walks      Years     CAtBat
         0          0          0          0
     CHits     CHmRun      CRuns       CRBI
         0          0          0          0
    CWalks    LeagueN  DivisionW    PutOuts
         0          0          0          0
   Assists     Errors NewLeagueN
         0          0          0
```

Hide

```
modellasso$lambda[70]
```

```
[1] 0.4310623
```

Hide

```
modellasso$a0[70]
```

```
     s69
264.2326
```

Hide

```
modellasso$beta[,70]
```

```
      AtBat         Hits        HmRun
 -0.5434582    0.0000000    8.1292599
       Runs          RBI        Walks
  0.0000000    0.1057942    4.2480742
      Years       CAtBat        CHits
-17.7360789   -0.4916314    2.2290690
     CHmRun        CRuns         CRBI
  1.9518105   -0.3334579    0.0000000
     CWalks      LeagueN    DivisionW
  0.1652172  102.0714817 -145.2585272
    PutOuts      Assists       Errors
  0.1885423    0.6043276   -3.6597910
 NewLeagueN
-48.8394407
```

Hide

```
modellasso$lambda[100]
```

```
[1] 0.02644958
```

Hide

```
modellasso$a0[100]
```

```
     s99
274.4948
```

Hide

```
modellasso$beta[,100]
```

```
      AtBat         Hits        HmRun
 -0.4100302   -1.2665960    6.3852223
       Runs          RBI        Walks
  1.2756560    0.8932321    3.8284479
      Years       CAtBat        CHits
-16.9892716   -0.6082740    2.9717990
     CHmRun        CRuns         CRBI
  3.0677219   -0.8696309   -0.4612084
     CWalks      LeagueN    DivisionW
  0.3122121  115.8333379 -144.8891401
    PutOuts      Assists       Errors
  0.1957103    0.6683825   -4.5586077
 NewLeagueN
-66.8223277
```

We can also return the coefficients using the coef command. It returns it for specific values of lambda. For values in the grid it returns the exact beta parameters while for values not in the grid, it returns values using interpolation between closest lambda values in the grid. You can use the exact=T with the training data to get the exact beta values.

Hide

```
coef(modellasso,s=15)
```

```
20 x 1 sparse Matrix of class "dgCMatrix"
                     s=15
(Intercept)   93.82856496
AtBat          .
Hits           0.29646857
HmRun          2.34348494
Runs           .
RBI            .
Walks          4.11083648
Years         -0.65173068
CAtBat         .
CHits          0.10247392
CHmRun         1.13708907
CRuns          .
CRBI           0.34482924
CWalks         .
LeagueN       34.57411925
DivisionW   -123.77714349
PutOuts        0.10005119
Assists        0.09653002
Errors         .
NewLeagueN     .
```

Hide

```
coef(modellasso,s=15,exact=T,x=X[train,],y=y[train])
```

```
20 x 1 sparse Matrix of class "dgCMatrix"
                     s=15
(Intercept)   92.41848657
AtBat          .
Hits           0.30496059
HmRun          2.33686628
Runs           .
RBI            .
Walks          4.11219805
Years         -0.32925757
CAtBat         .
CHits          0.09898385
CHmRun         1.13566970
CRuns          .
CRBI           0.34694553
CWalks         .
LeagueN       34.60588585
DivisionW   -123.75235053
PutOuts        0.09986349
Assists        0.09655562
Errors         .
NewLeagueN     .
```

Predictions: We start with a prediction for the model fitted with lambda = 100. The test mean squared error for this model is 151898.1. Suppose, we change lambda to 50, we get 144989 and if we change lambda to 200, we get 183137.7. Note that by default if prediction is done at lambda values that are not tried in the fitting algorithm, it uses linear interpolation to make predictions. We can use exact=T in the argument to get the exact value by refitting. In addition, you need to then pass also the original training set data to the function. You get a test error of 167213.2 with the full model (all variables) while 224669.9 with a very large value of lambda. Thus choosing lambda appropriately will be important in the quality of the fit. This can be done with cross-validation.

Hide

```
modellasso$lambda
```

```
  [1] 264.49580153 240.99872350 219.58906113
  [4] 200.08137415 182.30669632 166.11107187
  [7] 151.35422207 137.90833013 125.65693417
 [10] 114.49391847 104.32259431  95.05486256
 [13]  86.61045055  78.91621683  71.90551763
 [16]  65.51762962  59.69722398  54.39388714
 [19]  49.56168412  45.15876070  41.14698086
 [22]  37.49159649  34.16094639  31.12618206
 [25]  28.36101783  25.84150318  23.54581526
 [28]  21.45406992  19.54814948  17.81154576
 [31]  16.22921712  14.78745820  13.47378117
 [34]  12.27680759  11.18616984  10.19242134
 [37]   9.28695472   8.46192726   7.71019296
 [40]   7.02524067   6.40113765   5.83247823
 [43]   5.31433693   4.84222589   4.41205589
 [46]   4.02010100   3.66296631   3.33755848
 [49]   3.04105898   2.77089968   2.52474059
 [52]   2.30044960   2.09608399   1.90987367
 [55]   1.74020576   1.58561067   1.44474939
 [58]   1.31640184   1.19945633   1.09289993
 [61]   0.99580971   0.90734472   0.82673872
 [64]   0.75329353   0.68637301   0.62539752
 [67]   0.56983893   0.51921601   0.47309029
 [70]   0.43106225   0.39276787   0.35787546
 [73]   0.32608279   0.29711449   0.27071966
 [76]   0.24666967   0.22475622   0.20478949
 [79]   0.18659656   0.17001984   0.15491574
 [82]   0.14115345   0.12861377   0.11718808
 [85]   0.10677741   0.09729160   0.08864848
 [88]   0.08077320   0.07359753   0.06705933
 [91]   0.06110196   0.05567383   0.05072792
 [94]   0.04622139   0.04211521   0.03837381
 [97]   0.03496479   0.03185861   0.02902838
[100]   0.02644958
```

Hide

```
predictlasso1 <- predict(modellasso,newx=X[test,],s=100)
mean((predictlasso1-y[test])^2)
```

```
[1] 151898.1
```

Hide

```
predictlasso2 <- predict(modellasso,newx=X[test,],s=50)
mean((predictlasso2-y[test])^2)
```

```
[1] 144989
```

Hide

```
predictlasso3 <- predict(modellasso,newx=X[test,],s=200)
mean((predictlasso3-y[test])^2)
```

```
[1] 183137.7
```

Hide

```
predictlasso1a <- predict(modellasso,newx=X[test,],s=100,exact=T,x=X[train,],y=y[train])
mean((predictlasso1a-y[test])^2)
```

```
[1] 151899.3
```

Hide

```
predictlasso2a <- predict(modellasso,newx=X[test,],s=50,exact=T,x=X[train,],y=y[train])
mean((predictlasso2a-y[test])^2)
```

```
[1] 145007
```

Hide

```
predictlasso3a <- predict(modellasso,newx=X[test,],s=200,exact=T,x=X[train,],y=y[train])
mean((predictlasso3a-y[test])^2)
```

```
[1] 183127.9
```

Hide

```
predictlasso4 <- predict(modellasso,newx=X[test,],s=0,exact=T,x=X[train,],y=y[train])
mean((predictlasso4-y[test])^2)
```

```
[1] 167213.2
```

Hide

```
predictlasso5 <- predict(modellasso,newx=X[test,],s=10^10,exact=T,x=X[train,],y=y[train])
mean((predictlasso5-y[test])^2)
```

```
[1] 224669.9
```

Cross-validation: By default, you perform 10 fold cross validation. Note that glmnet uses randomization in choosing the folds which we should be able to control better by setting the seed to be the same. The optimal value of lambda found from cross validation is 8.461927. You can plot the relation between the lambda parameter and the cross-validated mean error (cvm). We see that the best fit from model with the optimal lambda gives a much smaller error on the test set (143803.3) than the model which is based on complete linear regression or the model with only an intercept. We can print out the coefficients to identify that 10 predictor variables are chosen (excluding the intercept).

Hide

```
set.seed(2)
cvlasso <- cv.glmnet(X[train,],y[train])
cvlasso$glmnet.fit
```

```
Call:  glmnet(x = X[train, ], y = y[train])

    Df  %Dev  Lambda
1    0  0.00 264.500
2    1  6.57 241.000
3    1 12.03 219.600
4    2 16.64 200.100
5    2 20.51 182.300
6    3 24.00 166.100
7    3 27.88 151.400
8    3 31.11 137.900
9    3 33.78 125.700
10   3 36.00 114.500
11   3 37.85 104.300
12   3 39.38  95.050
13   4 40.72  86.610
14   4 42.28  78.920
15   6 43.60  71.910
16   6 44.73  65.520
17   6 45.67  59.700
18   7 46.55  54.390
19   8 47.33  49.560
20   8 48.00  45.160
21   8 48.55  41.150
22  10 49.02  37.490
23   8 49.43  34.160
24   8 49.78  31.130
25   9 50.14  28.360
26   9 50.47  25.840
27  10 50.81  23.550
28  10 51.10  21.450
29  10 51.35  19.550
30  10 51.55  17.810
31  10 51.72  16.230
32  11 51.95  14.790
33  11 52.36  13.470
34  11 52.70  12.280
35  11 52.99  11.190
36  10 53.21  10.190
37  10 53.38   9.287
38  10 53.52   8.462
39  10 53.63   7.710
40  11 53.74   7.025
41  11 53.87   6.401
42  12 54.09   5.832
43  12 54.28   5.314
44  12 54.44   4.842
45  12 54.57   4.412
46  13 54.73   4.020
47  13 54.91   3.663
48  13 55.06   3.338
49  13 55.19   3.041
50  13 55.29   2.771
51  14 55.44   2.525
```

```
52  14 55.87   2.300
53  14 56.25   2.096
54  15 56.57   1.910
55  15 56.86   1.740
56  15 57.10   1.586
57  15 57.30   1.445
58  15 57.47   1.316
59  16 57.61   1.199
60  17 57.73   1.093
61  17 57.85   0.996
62  17 57.96   0.907
63  17 58.05   0.827
64  18 58.14   0.753
65  17 58.20   0.686
66  17 58.25   0.625
67  16 58.29   0.570
68  16 58.32   0.519
69  16 58.34   0.473
70  16 58.36   0.431
71  16 58.38   0.393
72  17 58.40   0.358
73  17 58.41   0.326
74  17 58.42   0.297
75  19 58.44   0.271
76  19 58.45   0.247
77  19 58.49   0.225
78  19 58.49   0.205
79  19 58.52   0.187
80  19 58.53   0.170
81  19 58.54   0.155
82  19 58.55   0.141
83  19 58.56   0.129
84  19 58.57   0.117
85  19 58.58   0.107
86  19 58.58   0.097
87  19 58.59   0.089
88  19 58.59   0.081
89  19 58.59   0.074
90  19 58.60   0.067
91  19 58.60   0.061
92  19 58.60   0.056
93  19 58.60   0.051
94  19 58.60   0.046
95  19 58.61   0.042
96  19 58.61   0.038
97  19 58.61   0.035
98  19 58.61   0.032
99  19 58.61   0.029
100 19 58.61   0.026
```

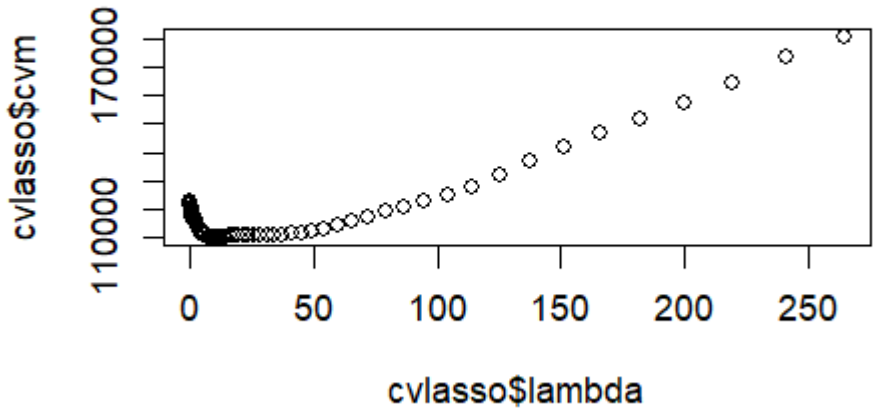
Hide

```
cvlasso$lambda.min
```

```
[1] 8.461927
```

Hide

```
plot(cvlasso$lambda,cvlasso$cvm)
```



Hide

```
predictlassocv <- predict(modellasso,s=8.461927,newx=X[test,])
mean((predictlassocv-y[test])^2)
```

```
[1] 143807.7
```

Hide

```
coef(modellasso,s=8.461927)
```

```
20 x 1 sparse Matrix of class "dgCMatrix"
               s=8.461927
(Intercept)  121.9078071
AtBat          .
Hits           .
HmRun          3.4527540
Runs           .
RBI            .
Walks          4.0095547
Years        -13.2996318
CAtBat         .
CHits          0.2172280
CHmRun         0.8938979
CRuns          .
CRBI           0.3905921
CWalks         .
LeagueN       49.0732189
DivisionW   -136.8585423
PutOuts        0.1182314
Assists        0.1424964
Errors         .
NewLeagueN     .
```

Hide

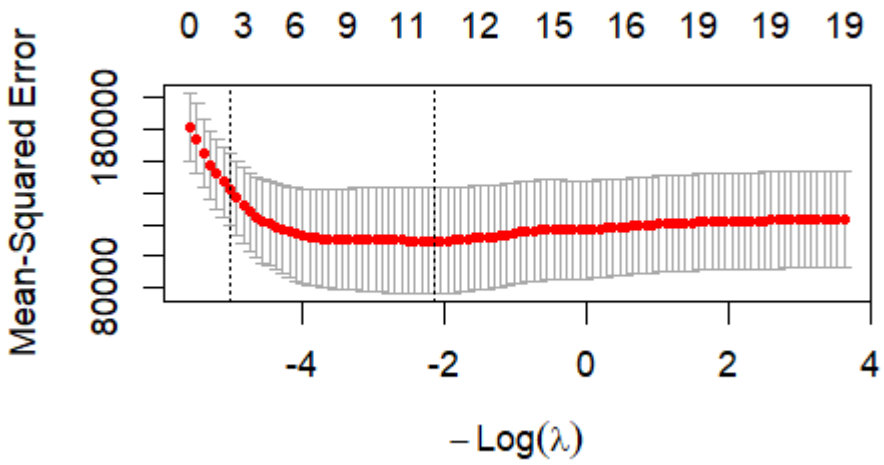
```
predictlassocvexact <- predict(modellasso,s=8.461927,newx=X[test,],exact=T,x=X[train,],y=y[train])
mean((predictlassocvexact-y[test])^2)
```

```
[1] 143803.3
```

You can also plot the output (cross-valiation error curve) directly for LASSO as follows:

Hide

```
plot(cvlasso)
```



Here's what each element means:

X-axis (bottom): −log(λ) — as you move left, λ increases (more regularization, fewer variables). Moving right means less regularization.

X-axis (top): The number of non-zero coefficients in the model at each λ value.

Y-axis: Cross-validated mean-squared error (CV MSE), averaged across the 10 folds.

Red dots: The CV MSE at each λ tried during fitting.

Gray bars: ±1 standard error around each CV MSE estimate, reflecting variability across folds.

Left dashed line (lambda.min): The λ that minimizes CV MSE — this is the best-fitting model (λ ≈ 8.46, 10 predictors in our case).

Right dashed line (lambda.1se): The largest λ thathose CV MSE is within 1 standard error of the minimum — a more parsimonious model that trades a small increase in error for fewer predictors (λ ≈ 151.35, 3 predictors in our case).

The wide gray bands indicate there's meaningful variability in the CV error, so the difference between models in the flat region of the curve is not large.

In the implementation of LASSO in glmnet the model that is solved is given by:

$$\min \frac{1}{2n} \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_1 - \ldots - \beta_p x_p)^2 + \lambda \sum_{j=1}^{p} |x_j|$$

and scale (standardize) the variable before running the fit. The coefficients are returned on the original scale. You can look at https://glmnet.stanford.edu/articles/glmnet.html (https://glmnet.stanford.edu/articles/glmnet.html) for details.